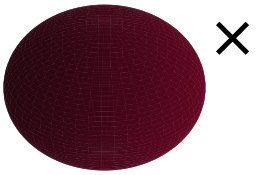
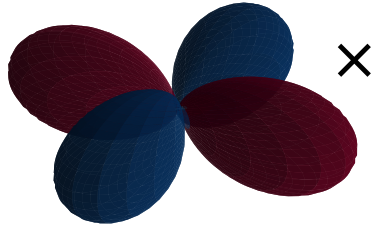


Wigner-Eckart Selection Rules: Overlap Integrals for $\hat{h}_+$ (Quadrupole Operator)

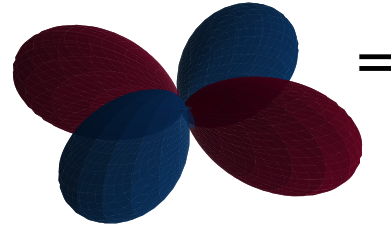
Initial $|s\rangle$ ($l = 0$)



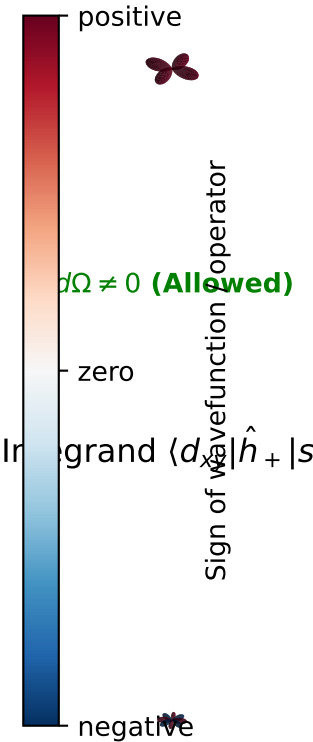
Operator $\hat{h}_+$ (rank-2)



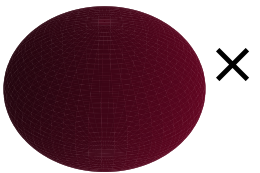
Final $\langle d_{x^2-y^2} |$ ($l = 2, m = \pm 2$)



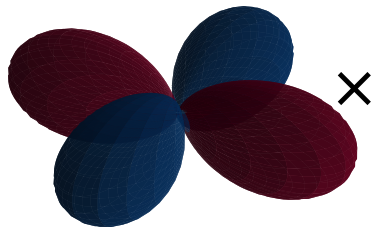
Integrand $\langle d_{x^2-y^2} | \hat{h}_+ | s \rangle$



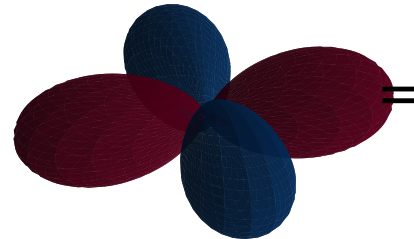
Initial $|s\rangle$ ($l = 0$)



Operator $\hat{h}_+$ (rank-2)



Final $\langle d_{xy} |$ ($l = 2, m = \pm 2?$)



Integrand $\langle d_{xy} | \hat{h}_+ | s \rangle$

$\int d\Omega = 0$ (Forbidden)